

NIKKALUOKTA, Sweden

WMO: 020360

Lat: 67.8527N

Lon: 19.0212E

Elev: 468

StdP: 95.83

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.6	(c) -34.0	(d) -40.5	(e) 0.1	(f) -36.4	(g) -37.6	(h) 0.1	(i) -33.9	(j) 12.3	(k) 1.8	(l) 11.0	(m) -0.1	(n) 0.9	(o) 310	(p) 0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	21.9	14.0	19.9	13.2	18.0	12.4	15.4	19.7	14.3	18.2	13.3	16.9	2.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	10.3	16.5	12.6	9.6	15.7	11.6	9.0	14.8	44.6	19.8	41.6	18.1	38.9	16.8	19.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.2	8.5	7.2	-39.5	24.4	2.5	2.1	-41.3	25.9	-42.8	27.1	-44.2	28.3	-46.0	29.8
			-39.6	16.8	2.3	1.0	-41.2	17.5	-42.6	18.1	-43.9	18.6	-45.6	19.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.0	-15.0	-14.5	-9.8	-3.0	3.7	9.1	12.3	10.4	5.8	-1.3	-9.9	-12.5		
	DBStd	11.59	9.48	9.63	7.25	4.94	3.80	3.36	2.70	3.03	3.44	5.44	7.85	8.80		
	HDD10.0	4538	774	686	613	391	198	56	8	32	132	352	598	698		
	HDD18.3	7416	1032	919	872	641	453	277	186	245	376	610	848	956		
	CDD10.0	164	0	0	0	0	3	29	81	45	6	0	0	0		
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH23.3	21	0	0	0	0	1	4	13	3	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0		
(14) Wind	WSAvg	2.6	2.3	2.6	3.0	2.6	3.2	3.1	2.5	2.3	2.6	2.5	1.9	2.4		
(15) Precipitation	PrecAvg	499	33	29	23	22	37	61	91	57	48	39	33	31		
	PrecMax	683	68	59	55	76	94	142	227	146	100	100	77	86		
	PrecMin	282	3	3	5	1	6	12	23	11	16	0	0	10		
	PrecStd	105	17	15	13	16	22	29	44	33	26	22	19	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	5.6	6.3	11.4	21.2	23.1	25.2	23.2	17.4	11.0	7.3	6.4		
		MCWB	2.6	2.6	2.6	6.1	10.8	13.4	15.4	15.2	11.8	7.9	4.5	3.2		
	2%	DB	3.9	3.7	4.2	7.5	16.4	20.3	22.3	20.9	14.8	9.3	4.9	4.1		
		MCWB	1.5	1.1	1.4	3.7	9.1	11.8	14.7	14.3	10.7	6.6	2.6	1.6		
	5%	DB	1.3	2.0	2.6	5.8	12.6	18.1	20.0	18.8	13.3	8.0	2.7	1.6		
		MCWB	-0.6	-0.1	0.2	2.5	7.4	11.2	13.9	13.8	9.8	5.5	1.0	-0.4		
	10%	DB	-1.2	-0.6	0.6	4.4	10.2	16.0	18.2	16.7	11.9	6.2	0.7	-0.3		
		MCWB	-2.8	-2.3	-1.4	1.5	5.9	10.4	12.8	12.5	8.9	4.0	-0.5	-1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	3.1	2.9	6.2	11.5	15.0	17.0	16.7	12.6	8.2	4.6	3.5		
		MCDB	5.4	5.5	5.6	11.5	19.6	20.7	22.2	21.3	15.6	10.3	7.1	6.0		
	2%	WB	1.5	1.4	1.6	4.0	9.5	13.2	15.7	15.2	11.4	6.8	2.9	1.7		
		MCDB	3.8	3.5	4.1	7.0	15.5	18.4	20.1	19.3	14.1	9.0	4.9	4.0		
	5%	WB	-0.5	0.0	0.3	2.8	7.7	11.9	14.7	14.1	10.3	5.5	1.1	-0.4		
		MCDB	1.3	2.1	2.5	5.4	12.6	16.6	18.7	18.0	12.7	7.8	2.4	1.4		
	10%	WB	-2.8	-2.1	-1.3	1.7	6.2	10.7	13.8	13.0	9.3	4.1	-0.4	-1.9		
		MCDB	-1.2	-0.5	0.6	4.1	9.7	15.1	17.2	16.2	11.5	6.0	0.7	-0.3		
(35) Mean Daily Temperature Range	MDBR		10.4	11.2	12.1	10.2	8.0	8.4	8.7	9.4	8.4	7.1	8.7	10.2		
	5% DB	MCDBR	10.0	10.6	9.7	10.6	13.7	13.2	13.3	14.0	11.4	7.9	9.4	9.8		
		MCWBR	8.6	8.7	8.0	7.5	7.8	6.5	6.5	8.1	7.5	5.9	7.7	8.6		
	5% WB	MCDBR	9.8	10.5	9.2	9.5	12.9	11.5	11.1	12.4	9.4	7.1	8.5	9.5		
		MCWBR	8.7	9.0	7.7	6.7	7.5	6.1	6.1	7.5	6.5	5.5	7.2	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.173	0.233	0.275	0.302	0.320	0.324	0.332	0.319	0.287	0.251	0.177	N/A		
	taud		1.921	2.117	2.187	2.228	2.363	2.466	2.493	2.510	2.563	2.284	1.962	N/A		
	Ebn at Noon		138	586	766	848	874	881	861	835	783	567	119	0		
	Edh at Noon		9	46	78	101	99	93	88	77	56	38	8	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.47	1.89	3.85	4.95	4.97	4.58	3.34	1.77	0.59	0.10	0.00		
	RadStd		0.00	0.05	0.11	0.18	0.38	0.41	0.46	0.31	0.19	0.09	0.01	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page